

# (MATH000) Mathematical Language

*Notation and Conventions*

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## Contents

|                                               |          |
|-----------------------------------------------|----------|
| <b>Part I. General Mathematical Language</b>  | <b>1</b> |
| 1 General Conventions                         | 1        |
| 2 Theorem-like Environments                   | 2        |
| <b>Part II. Notation by Mathematical Area</b> | <b>5</b> |
| 3 Set Theory and Logic                        | 5        |
| 4 Calculus and Analysis                       | 6        |
| 5 Linear Algebra                              | 9        |
| 6 Differential Equations                      | 10       |
| 7 Combinatorics and Discrete Mathematics      | 12       |
| 8 Probability and Statistics                  | 12       |
| 9 Geometry                                    | 14       |
| 10 Number Theory                              | 14       |
| 11 Algebra                                    | 15       |
| 12 Topology                                   | 17       |
| 13 Miscellaneous Notation                     | 17       |

## Part I. General Mathematical Language

### 1 General Conventions

These conventions apply throughout. Widely used standard alternatives may be listed for the same concept in preferred order; the first is used throughout.

| Item                       | Convention                                                                                                                                                                                                       |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scalars                    | Ordinary mathematical italic letters, for example $a, b, c, \lambda, t$ .                                                                                                                                        |
| Vectors                    | Bold lowercase letters, for example $\mathbf{u}, \mathbf{v}, \mathbf{x}$ . Unless otherwise stated, vectors are column vectors.                                                                                  |
| Matrices                   | Bold uppercase letters, for example $\mathbf{A}, \mathbf{B}, \mathbf{M}$ . Parentheses are the preferred matrix delimiters; square brackets are a standard alternative.                                          |
| Greek vectors and matrices | Bold Greek letters when a Greek symbol denotes a vector or matrix, for example $\boldsymbol{\xi}$ and $\boldsymbol{\Phi}$ .                                                                                      |
| Dimension subscripts       | Prefer dimension subscripts when the dimension is readily available and useful; omit them when it is clear, immaterial, or not readily available rather than inferring a dimension solely to supply a subscript. |
| Number systems             | Blackboard bold, for example $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ .                                                                                                                      |
| Sets                       | Ordinary uppercase letters such as $A, B, X, Y$ , unless a field-specific convention requires otherwise.                                                                                                         |
| Families and collections   | Calligraphic letters when useful, for example $\mathcal{F}, \mathcal{T}, \mathcal{B}$ .                                                                                                                          |
| Named operators            | Upright Roman type, for example $\det, \ker, \operatorname{im}, \operatorname{rk}$ , and $\operatorname{tr}$ .                                                                                                   |
| Large operators            | Display size in displayed mathematics and notation tables; sums and products place indices above and below, limits place subscripts below, and integral bounds retain standard right-side placement.             |
| Greek variants             | Distinct glyphs represent distinct symbols; in particular, $\varepsilon$ and $\epsilon$ are not interchangeable.                                                                                                 |

| Notation                             | Meaning                                                                                                                                 |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| $(a_1, \dots, a_n)$                  | Ordered tuple.                                                                                                                          |
| $(a_n)_{n \in I}, \{a_n\}_{n \in I}$ | Sequence indexed by $I$ ; parentheses are preferred, while braces are a standard alternative when the sequence interpretation is clear. |
| $(a, b)$                             | Open interval.                                                                                                                          |
| $[a, b]$                             | Closed interval.                                                                                                                        |
| $[a, b)$                             | Left-closed, right-open interval.                                                                                                       |
| $(a, b]$                             | Left-open, right-closed interval.                                                                                                       |
| $\{a_1, \dots, a_n\}$                | Set delimiters.                                                                                                                         |
| $\{x \in A \mid P(x)\}$              | Set-builder notation.                                                                                                                   |
| $\langle u, v \rangle$               | Pairing or inner product, according to context.                                                                                         |
| $ x $                                | Absolute value, modulus, cardinality, or order, according to context.                                                                   |
| $\ x\ $                              | Norm.                                                                                                                                   |
| s.t.                                 | Abbreviation for “such that” when naturally required.                                                                                   |
| $a = b$                              | Equality.                                                                                                                               |
| $a := b$                             | $a$ is defined to be $b$ .                                                                                                              |
| $a \equiv b$                         | Identity, equivalence, or congruence, according to context.                                                                             |
| $a \cong b$                          | Isomorphism or congruence, according to context.                                                                                        |
| $a \approx b$                        | Approximation.                                                                                                                          |
| $f: X \rightarrow Y$                 | Function or map from $X$ to $Y$ .                                                                                                       |
| $x \mapsto f(x)$                     | Elementwise rule of a map.                                                                                                              |

| Notation                    | Meaning                             |
|-----------------------------|-------------------------------------|
| $g \circ f$                 | Composition: first $f$ , then $g$ . |
| $f _A, f \upharpoonright_A$ | Restriction of $f$ to $A$ .         |
| $\text{id}_X$               | Identity map on $X$ .               |

## 2 Theorem-like Environments

| Item           | Convention                                                                   |
|----------------|------------------------------------------------------------------------------|
| Definition     | Precise meaning of a term, object, property, or construction.                |
| Notation       | Introduction or declaration of notation.                                     |
| Theorem        | Principal mathematical result.                                               |
| Proposition    | Useful result of more limited scope.                                         |
| Lemma          | Supporting result, usually for a later proof.                                |
| Corollary      | Result following readily from a preceding result.                            |
| Conjecture     | Statement proposed to be true but not proved in full generality.             |
| Hypothesis     | Named assumption or proposed statement.                                      |
| Formula        | Distinguished computational identity or closed form.                         |
| Algorithm      | Explicit step-by-step procedure.                                             |
| Fact           | Verified statement not requiring theorem status.                             |
| Example        | Illustration of a definition, construction, phenomenon, or result.           |
| Counterexample | Example disproving a general assertion or showing a hypothesis is necessary. |
| Remark         | Supplementary context, warning, or qualification.                            |
| Problem        | Mathematical task intended to be solved.                                     |
| Open Problem   | Problem not known to be solved in the stated form.                           |
| Idea           | Decisive conceptual insight.                                                 |
| Strategy       | Broad problem-solving plan.                                                  |
| Tactic         | Local step within a larger strategy.                                         |

Proof labels are written as *Proof* and solution labels as *Solution*, each followed by a period.

| Notation | Meaning                                                                 |
|----------|-------------------------------------------------------------------------|
| □        | Ending for ordinary proofs and solutions.                               |
| ■        | Ending for substantial, difficult, or particularly important proofs.    |
| Q.E.D.   | Ending for exceptionally difficult, significant, or paper-level proofs. |

**Enumeration and Subdivision.** Enumeration labels are set in upright Roman type. Their form is determined by structural role rather than nesting depth.

| Item                  | Convention                                                                                                                                                            |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1, 1.2, 1.3, ...    | Independent child problems or standalone units under a common parent; each may be referenced separately.                                                              |
| (1), (2), (3), ...    | Ordinary enumerated items, such as choices, cases, examples, or procedural steps.                                                                                     |
| (a), (b), (c), ...    | Connected subparts of a single problem sharing a common setup.                                                                                                        |
| (i), (ii), (iii), ... | Formal mathematical clauses, such as hypotheses, conditions, properties, or equivalent statements within definitions, theorems, propositions, and related statements. |

The distinction between 1.1 and (a) is independence: 1.1 denotes a standalone child unit, whereas (a) denotes a connected subpart of one problem. Likewise, (1) is used for ordinary enumeration, while (i) is reserved for formal mathematical clauses. Nested enumeration preserves these roles; for example, a subpart (a) may contain formal clauses (i), (ii), . . . .

**Mathematical Shorthand.** Typeset abbreviations may be used in typeset mathematical writing as well as handwritten work. Handwritten shorthand is reserved for notes, scratch work, and blackboard writing.

*Typeset Abbreviations.*

| Notation | Meaning                                                                                 |
|----------|-----------------------------------------------------------------------------------------|
| i.e.     | “that is.”                                                                              |
| i.o.w.   | “in other words.”                                                                       |
| s.t.     | “such that”; preferred when an abbreviation is natural in typeset mathematical writing. |
| iff      | “if and only if.”                                                                       |
| w.r.t.   | “with respect to.”                                                                      |
| const    | “constant.”                                                                             |
| N.t.     | “note that.”                                                                            |
| w.l.o.g. | “without loss of generality.”                                                           |
| w.t.s.   | “want to show (that).”                                                                  |
| e.t.s.   | “enough to show (that).”                                                                |
| t.f.a.e. | “the following are equivalent.”                                                         |
| i.t.s.w. | “in the same way.”                                                                      |
| o.t.o.h. | “on the other hand.”                                                                    |
| IVT      | Intermediate Value Theorem.                                                             |
| EVT      | Extreme Value Theorem.                                                                  |
| MVT      | Mean Value Theorem.                                                                     |
| FTC      | Fundamental Theorem of Calculus.                                                        |
| ODE      | Ordinary Differential Equation.                                                         |
| PDE      | Partial Differential Equation.                                                          |
| IVP      | Initial Value Problem.                                                                  |
| BVP      | Boundary Value Problem.                                                                 |

*Handwritten Shorthand.*

| Notation | Meaning           |
|----------|-------------------|
| Def      | “definition.”     |
| Thm      | “theorem.”        |
| Prop     | “proposition.”    |
| Lem      | “lemma.”          |
| Cor      | “corollary.”      |
| Pf       | “proof.”          |
| Sol      | “solution.”       |
| Ex       | “example.”        |
| Exer     | “exercise.”       |
| Rmk      | “remark.”         |
| Sps      | “suppose (that).” |
| cont     | “continuous.”     |
| cont.ly  | “continuously.”   |

| Notation | Meaning             |
|----------|---------------------|
| difb     | “differentiable.”   |
| difb.ly  | “differentiably.”   |
| difl     | “differential.”     |
| bdd      | “bounded.”          |
| unif     | “uniform.”          |
| unif.ly  | “uniformly.”        |
| lin      | “linear.”           |
| lin.ly   | “linearly.”         |
| dep      | “dependent.”        |
| indep    | “independent.”      |
| approx   | “approximate.”      |
| triv     | “trivial.”          |
| fnc      | “function.”         |
| seq      | “sequence.”         |
| eqn      | “equation.”         |
| eqlt     | “equality.”         |
| ineqlt   | “inequality.”       |
| sys      | “system.”           |
| comb     | “combination.”      |
| vec      | “vector.”           |
| mat      | “matrix.”           |
| col      | “column.”           |
| pt       | “point.”            |
| bdy      | “boundary.”         |
| nbhd     | “neighborhood.”     |
| itvl     | “interval.”         |
| op       | “operation.”        |
| num      | “number.”           |
| val      | “value.”            |
| elmt     | “element.”          |
| cw       | “clockwise.”        |
| ccw      | “counterclockwise.” |
| w/       | “with.”             |
| w/o      | “without.”          |

**Mathematical Phrases.**

| Phrase                         | Use                                                      |
|--------------------------------|----------------------------------------------------------|
| We find by inspection that ... | Used when a conclusion follows directly from inspection. |

## Part II. Notation by Mathematical Area

### 3 Set Theory and Logic

Uppercase letters usually denote sets; lowercase letters usually denote their elements.

| Notation                               | Meaning                                                                                     |
|----------------------------------------|---------------------------------------------------------------------------------------------|
| $\mathbb{N} = \{1, 2, 3, \dots\}$      | Positive integers.                                                                          |
| $\mathbb{N}_0 = \{0, 1, 2, 3, \dots\}$ | Nonnegative integers.                                                                       |
| $\mathbb{Z}$                           | Integers.                                                                                   |
| $\mathbb{Q}$                           | Rational numbers.                                                                           |
| $\mathbb{R}$                           | Real numbers.                                                                               |
| $\mathbb{C}$                           | Complex numbers.                                                                            |
| $\emptyset, \emptyset$                 | Empty set.                                                                                  |
| $x \in A$                              | $x$ is an element of $A$ .                                                                  |
| $x \notin A$                           | $x$ is not an element of $A$ .                                                              |
| $A \subseteq B$                        | $A$ is a subset of $B$ ; equality is allowed.                                               |
| $A \subsetneq B$                       | $A$ is a proper subset of $B$ .                                                             |
| $A \supseteq B$                        | $A$ is a superset of $B$ ; equality is allowed.                                             |
| $A \supsetneq B$                       | $A$ is a proper superset of $B$ .                                                           |
| $A \cup B$                             | Union.                                                                                      |
| $A \cap B$                             | Intersection.                                                                               |
| $\bigcup_{i \in I} A_i$                | Union of an indexed family of sets.                                                         |
| $\bigcap_{i \in I} A_i$                | Intersection of an indexed family of sets.                                                  |
| $A \setminus B, A - B$                 | Set difference.                                                                             |
| $A \Delta B, A \triangle B$            | Symmetric difference.                                                                       |
| $A^c$                                  | Complement of $A$ in the ambient set.                                                       |
| $\mathcal{P}(A), 2^A$                  | Power set of $A$ .                                                                          |
| $A \times B$                           | Cartesian product.                                                                          |
| $A \sqcup B, A \dot{\cup} B$           | Disjoint union.                                                                             |
| $A \cap B = \emptyset$                 | $A$ and $B$ are disjoint.                                                                   |
| $ A , \#A$                             | Cardinality.                                                                                |
| $\{x \in A \mid P(x)\}$                | Elements $x \in A$ satisfying $P(x)$ .                                                      |
| $R \subseteq X \times Y$               | Binary relation between $X$ and $Y$ .                                                       |
| $xRy$                                  | $x$ is related to $y$ by the relation $R$ .                                                 |
| $R^{-1}$                               | Inverse relation of $R$ .                                                                   |
| $[x]_R, [x], R[x]$                     | Equivalence class of $x$ under $R$ ; abbreviated forms are used when the relation is clear. |
| $x \sim y$                             | Equivalent elements under an understood equivalence relation.                               |
| $A/\sim$                               | Set of equivalence classes.                                                                 |
| $\forall x \in A$                      | For every $x \in A$ .                                                                       |
| $\exists x \in A$                      | There exists $x \in A$ .                                                                    |
| $\exists! x \in A$                     | There exists a unique $x \in A$ .                                                           |
| $\nexists x \in A$                     | There does not exist $x \in A$ .                                                            |
| $P \Rightarrow Q$                      | $P$ implies $Q$ .                                                                           |
| $P \Leftarrow Q$                       | $Q$ implies $P$ .                                                                           |
| $P \nRightarrow Q$                     | $P$ does not imply $Q$ .                                                                    |
| $P \Leftrightarrow Q$                  | $P$ if and only if $Q$ .                                                                    |

| Notation                        | Meaning                                                      |
|---------------------------------|--------------------------------------------------------------|
| $P \wedge Q$                    | Logical conjunction.                                         |
| $P \vee Q$                      | Logical disjunction.                                         |
| $\neg P$                        | Logical negation.                                            |
| $\nmid$                         | Contradiction reached in a proof or argument.                |
| $\therefore$                    | Therefore; indicates a conclusion from preceding statements. |
| $\because$                      | Because; indicates a reason or justification.                |
| $\text{dom } f$                 | Domain of $f$ .                                              |
| $\text{codom } f$               | Codomain of $f$ .                                            |
| $f(A)$                          | Image of $A$ under $f$ .                                     |
| $f^{-1}(B)$                     | Preimage of $B$ under $f$ .                                  |
| $\text{im } f$                  | Image of the map $f$ .                                       |
| $f: X \hookrightarrow Y$        | Injective map.                                               |
| $f: X \twoheadrightarrow Y$     | Surjective map.                                              |
| $f: X \xleftrightarrow{\sim} Y$ | Bijjective map.                                              |
| $f: X \xrightarrow{\sim} Y$     | Isomorphism; for sets, equivalently a bijection.             |
| $\mathbf{1}_A(x)$               | Indicator function of $A$ .                                  |
| $\aleph_0$                      | Cardinality of a countably infinite set.                     |
| $\mathfrak{c}$                  | Cardinality of the continuum.                                |

## 4 Calculus and Analysis

The differential  $d$  is upright;  $\varepsilon$  is the default small positive quantity in analysis. Function spaces take their domains in parentheses, for example  $C([a, b])$ ,  $\mathcal{R}([a, b])$ , and  $L^p(\Omega)$ .

| Notation                        | Meaning                                          |
|---------------------------------|--------------------------------------------------|
| $\lim_{x \rightarrow a} f(x)$   | Limit as $x$ approaches $a$ .                    |
| $\lim_{x \rightarrow a^-} f(x)$ | Left-hand limit as $x$ approaches $a$ .          |
| $\lim_{x \rightarrow a^+} f(x)$ | Right-hand limit as $x$ approaches $a$ .         |
| $\min A$                        | Minimum of $A$ .                                 |
| $\max A$                        | Maximum of $A$ .                                 |
| $\inf A$                        | Infimum of $A$ .                                 |
| $\sup A$                        | Supremum of $A$ .                                |
| $f'(a)$                         | First derivative of $f$ at $a$ .                 |
| $f^{(n)}(a)$                    | $n$ th derivative of $f$ at $a$ .                |
| $\frac{dy}{dx}$                 | Derivative of $y$ with respect to $x$ .          |
| $\frac{dx}{df}, \partial_i f$   | Partial derivative with respect to $x_i$ .       |
| $\int_a^b f(x) \, dx$           | Definite integral over $[a, b]$ .                |
| $\iint_D f \, dA$               | Double integral over $D$ .                       |
| $\iiint_\Omega f \, dV$         | Triple integral over $\Omega$ .                  |
| $\int_C f(z) \, dz$             | Contour integral along an oriented contour $C$ . |
| $\oint_C f(z) \, dz$            | Contour integral over a closed contour $C$ .     |
| $-C$                            | The contour $C$ with the opposite orientation.   |

| Notation                                   | Meaning                                                                                        |
|--------------------------------------------|------------------------------------------------------------------------------------------------|
| $\gamma^{-1}$                              | Reversed parametrized path $\gamma$ , when unambiguous; not a functional inverse.              |
| PV, P.V.                                   | Principal value; in particular, the Cauchy principal value of a singular or improper integral. |
| $\sum_{k=1}^n a_k$                         | Finite sum.                                                                                    |
| $\sum_{k=1}^{\infty} a_k$                  | Infinite series.                                                                               |
| $\prod_{k=1}^n a_k$                        | Finite product.                                                                                |
| $e^x$ , $\exp x$                           | Exponential function.                                                                          |
| $\ln x$ , $\log_e x$                       | Logarithm to base $e$ .                                                                        |
| $\lg x$ , $\log_{10} x$                    | Logarithm to base 10.                                                                          |
| $\text{lb } x$ , $\log_2 x$                | Logarithm to base 2.                                                                           |
| $\log_a x$                                 | Logarithm to base $a$ .                                                                        |
| $\sin x$                                   | Sine.                                                                                          |
| $\cos x$                                   | Cosine.                                                                                        |
| $\tan x$                                   | Tangent.                                                                                       |
| $\csc x$                                   | Cosecant.                                                                                      |
| $\sec x$                                   | Secant.                                                                                        |
| $\cot x$                                   | Cotangent.                                                                                     |
| $\arcsin x$ , $\sin^{-1} x$                | Principal inverse sine.                                                                        |
| $\arccos x$ , $\cos^{-1} x$                | Principal inverse cosine.                                                                      |
| $\arctan x$ , $\tan^{-1} x$                | Principal inverse tangent.                                                                     |
| $\text{arccsc } x$ , $\csc^{-1} x$         | Principal inverse cosecant.                                                                    |
| $\text{arcsec } x$ , $\sec^{-1} x$         | Principal inverse secant.                                                                      |
| $\text{arccot } x$ , $\cot^{-1} x$         | Principal inverse cotangent.                                                                   |
| $\sinh x$                                  | Hyperbolic sine.                                                                               |
| $\cosh x$                                  | Hyperbolic cosine.                                                                             |
| $\tanh x$                                  | Hyperbolic tangent.                                                                            |
| $\text{csch } x$                           | Hyperbolic cosecant.                                                                           |
| $\text{sech } x$                           | Hyperbolic secant.                                                                             |
| $\coth x$                                  | Hyperbolic cotangent.                                                                          |
| $\text{arcsinh } x$ , $\sinh^{-1} x$       | Principal inverse hyperbolic sine.                                                             |
| $\text{arcosh } x$ , $\cosh^{-1} x$        | Principal inverse hyperbolic cosine.                                                           |
| $\text{artanh } x$ , $\tanh^{-1} x$        | Principal inverse hyperbolic tangent.                                                          |
| $\text{arccsch } x$ , $\text{csch}^{-1} x$ | Principal inverse hyperbolic cosecant.                                                         |
| $\text{arcsech } x$ , $\text{sech}^{-1} x$ | Principal inverse hyperbolic secant.                                                           |
| $\text{arcoth } x$ , $\coth^{-1} x$        | Principal inverse hyperbolic cotangent.                                                        |
| $\Gamma(z)$                                | Gamma function.                                                                                |
| $B(x, y)$                                  | Beta function.                                                                                 |
| $\zeta(s)$                                 | Riemann zeta function.                                                                         |
| $D\mathbf{F}(\mathbf{a})$                  | Total derivative; Jacobian matrix in coordinates.                                              |
| $\text{Jac}(\mathbf{F})$                   | Jacobian determinant.                                                                          |
| $\text{Hess } f$                           | Hessian matrix of $f$ .                                                                        |
| $\nabla f$                                 | Gradient of $f$ .                                                                              |

| Notation                                                            | Meaning                                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $\nabla \cdot \mathbf{F}$                                           | Divergence of $\mathbf{F}$ .                                                                        |
| $\nabla \times \mathbf{F}$                                          | Curl of $\mathbf{F}$ .                                                                              |
| $\Delta f$                                                          | Laplacian of $f$ .                                                                                  |
| $\mathbf{r}(t)$                                                     | Parametrized curve.                                                                                 |
| $ds$                                                                | Element of arc length.                                                                              |
| $\int_C \mathbf{F} \cdot d\mathbf{r}$                               | Line integral along $C$ .                                                                           |
| $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$                         | Flux integral through $S$ .                                                                         |
| $x_n \rightarrow x$                                                 | Convergence of a sequence.                                                                          |
| $a_n \nearrow L$                                                    | Monotone increasing convergence to $L$ .                                                            |
| $a_n \searrow L$                                                    | Monotone decreasing convergence to $L$ .                                                            |
| $\limsup a_n$                                                       | Limit superior.                                                                                     |
| $\liminf a_n$                                                       | Limit inferior.                                                                                     |
| $f_n \rightarrow f, f_n \xrightarrow{\text{ptw}} f$                 | Pointwise convergence.                                                                              |
| $f_n \rightrightarrows f, f_n \xrightarrow{\text{unif}} f$          | Uniform convergence.                                                                                |
| $f_n \xrightarrow{\text{a.e.}} f$                                   | Convergence almost everywhere.                                                                      |
| $C(\Omega), C^0(\Omega)$                                            | Continuous functions on $\Omega$ .                                                                  |
| $C^k(\Omega)$                                                       | $k$ times continuously differentiable functions on $\Omega$ .                                       |
| $C^\infty(\Omega)$                                                  | Smooth functions on $\Omega$ .                                                                      |
| $\mathcal{R}([a, b])$                                               | Riemann-integrable functions on $[a, b]$ .                                                          |
| $L^p(\Omega)$                                                       | Lebesgue $L^p$ space on $\Omega$ .                                                                  |
| $L^\infty(\Omega)$                                                  | Lebesgue space of essentially bounded functions on $\Omega$ .                                       |
| $\ f\ _\infty$                                                      | Supremum norm $\sup_{x \in \Omega}  f(x) $ , when finite.                                           |
| $\ f\ _{L^p(\Omega)}, \ f\ _p$                                      | $L^p$ -norm $\left( \int_\Omega  f(x) ^p \, dx \right)^{1/p}$ for $1 \leq p < \infty$ .             |
| $\ f\ _{L^\infty(\Omega)}$                                          | $L^\infty$ -norm $\text{ess sup}_{x \in \Omega}  f(x) $ .                                           |
| $B_r(x)$                                                            | Open ball of radius $r$ centered at $x$ .                                                           |
| $\overline{B}_r(x)$                                                 | Closed ball of radius $r$ centered at $x$ .                                                         |
| $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$                 | Extended complex plane, equivalently the Riemann sphere.                                            |
| $z = x + iy$                                                        | Complex number with real part $x$ and imaginary part $y$ .                                          |
| $\operatorname{Re} z, \Re z$                                        | Real part of $z$ .                                                                                  |
| $\operatorname{Im} z, \Im z$                                        | Imaginary part of $z$ ; distinct from lowercase im for image.                                       |
| $\bar{z}$                                                           | Complex conjugate.                                                                                  |
| $ z $                                                               | Complex modulus.                                                                                    |
| $\arg z$                                                            | Argument of a nonzero complex number $z$ ; its values differ by integer multiples of $2\pi$ .       |
| $\operatorname{Arg} z$                                              | Principal argument of $z \neq 0$ , with $-\pi < \operatorname{Arg} z \leq \pi$ .                    |
| $\log z$                                                            | Complex logarithm of $z \neq 0$ ; a branch is specified when a single-valued logarithm is required. |
| $\operatorname{Log} z = \ln  z  + i \operatorname{Arg} z$           | Principal complex logarithm for $z \neq 0$ .                                                        |
| $\operatorname{Res}(f; a), \operatorname{Res}_{z=a} f(z)$           | Residue of $f$ at a finite point $a$ .                                                              |
| $\operatorname{Res}(f; \infty), \operatorname{Res}_{z=\infty} f(z)$ | Residue of $f$ at infinity.                                                                         |

## 5 Linear Algebra

| Notation                                                              | Meaning                                                                                              |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| $\mathbf{A} = (a_{ij})$ , $\mathbf{A} = [a_{ij}]$                     | Matrix with $(i, j)$ -entry $a_{ij}$ ; parentheses are preferred.                                    |
| $\mathbf{A}^\top$                                                     | Transpose of $\mathbf{A}$ .                                                                          |
| $\mathbf{A}^H$                                                        | Conjugate transpose of $\mathbf{A}$ .                                                                |
| $\mathbf{A}^{-1}$                                                     | Inverse of $\mathbf{A}$ .                                                                            |
| $\mathbf{I}_n$ , $\mathbf{I}$                                         | Identity matrix; the dimension subscript is preferred when readily available and useful.             |
| $\mathbf{0}$                                                          | Zero vector or zero matrix, with dimension understood.                                               |
| $\mathbf{D}$                                                          | Diagonal matrix when a local name is required.                                                       |
| $\mathbf{P}$                                                          | Projection matrix when a local name is required.                                                     |
| $\text{diag}(d_1, \dots, d_n)$                                        | Diagonal matrix with entries $d_1, \dots, d_n$ .                                                     |
| $\mathbf{1}$                                                          | All-ones vector.                                                                                     |
| $e^{\mathbf{A}}$ , $\exp(\mathbf{A})$                                 | Matrix exponential; $e^{t\mathbf{A}}$ is the standard time-dependent form.                           |
| $\det(\mathbf{A})$ , $ \mathbf{A} $                                   | Determinant.                                                                                         |
| $\text{rk}(\mathbf{A})$ , $\text{rank}(\mathbf{A})$                   | Rank.                                                                                                |
| $\text{tr}(\mathbf{A})$                                               | Trace.                                                                                               |
| $\text{adj}(\mathbf{A})$                                              | Adjugate matrix.                                                                                     |
| $\text{Col}(\mathbf{A})$                                              | Column space.                                                                                        |
| $\text{Row}(\mathbf{A})$                                              | Row space.                                                                                           |
| $\text{Null}(\mathbf{A})$ , $\ker \mathbf{A}$                         | Null space.                                                                                          |
| $\ker T$                                                              | Kernel of a linear map $T$ .                                                                         |
| $\text{im } T$                                                        | Image of a linear map $T$ .                                                                          |
| $\mathbf{e}_i$                                                        | $i$ th standard basis vector.                                                                        |
| $\dim V$                                                              | Dimension of $V$ .                                                                                   |
| $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$                      | Span of the listed vectors.                                                                          |
| $U \oplus V$                                                          | Direct sum.                                                                                          |
| $\langle \mathbf{u}, \mathbf{v} \rangle$ , $(\mathbf{u}, \mathbf{v})$ | Inner product; over complex spaces, conjugate-linear in the first argument and linear in the second. |
| $\mathbf{u} \cdot \mathbf{v}$ , $\mathbf{u}^\top \mathbf{v}$          | Standard Euclidean inner product on $\mathbb{R}^n$ .                                                 |
| $\mathbf{u}^H \mathbf{v}$                                             | Standard Hermitian inner product on $\mathbb{C}^n$ .                                                 |
| $\ \mathbf{v}\ $                                                      | Norm of $\mathbf{v}$ .                                                                               |
| $\mathbf{u} \perp \mathbf{v}$                                         | Orthogonality.                                                                                       |
| $U^\perp$                                                             | Orthogonal complement of the subspace $U$ .                                                          |
| $\text{proj}_U \mathbf{x}$                                            | Orthogonal projection of $\mathbf{x}$ onto the subspace $U$ .                                        |
| $\text{proj}_{\mathbf{v}} \mathbf{x}$                                 | Orthogonal projection of $\mathbf{x}$ onto $\text{span}(\mathbf{v})$ .                               |
| $\mathbf{u} \times \mathbf{v}$                                        | Cross product in $\mathbb{R}^3$ .                                                                    |
| $p_{\mathbf{A}}(\lambda)$ , $\chi_{\mathbf{A}}(\lambda)$              | Characteristic polynomial of $\mathbf{A}$ .                                                          |
| $m_{\mathbf{A}}(\lambda)$ , $\mu_{\mathbf{A}}(\lambda)$               | Minimal polynomial of $\mathbf{A}$ .                                                                 |
| $\lambda$ , $\lambda_i$                                               | Eigenvalue of $\mathbf{A}$ .                                                                         |
| $\mathbf{v}$ , $\mathbf{v}_i$                                         | Eigenvector of $\mathbf{A}$ .                                                                        |
| $E_\lambda(\mathbf{A})$ , $V_\lambda(\mathbf{A})$                     | Eigenspace of $\mathbf{A}$ associated with $\lambda$ .                                               |
| $\sigma(\mathbf{A})$ , $\text{spec}(\mathbf{A})$                      | Spectrum of $\mathbf{A}$ .                                                                           |
| $\rho(\mathbf{A})$                                                    | Spectral radius.                                                                                     |
| $\sigma_i(\mathbf{A})$                                                | $i$ th singular value of $\mathbf{A}$ .                                                              |

## 6 Differential Equations

| Notation                                                                                                           | Meaning                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| $y'$                                                                                                               | First derivative with respect to the understood independent variable.                                                                      |
| $y''$                                                                                                              | Second derivative.                                                                                                                         |
| $\dot{x}$                                                                                                          | First derivative of $x$ with respect to time.                                                                                              |
| $\ddot{x}$                                                                                                         | Second derivative of $x$ with respect to time.                                                                                             |
| $F(t, y, y', \dots, y^{(n)}) = 0$                                                                                  | General $n$ th-order ordinary differential equation.                                                                                       |
| $y^{(n)} = f(t, y, \dots, y^{(n-1)})$                                                                              | $n$ th-order ordinary differential equation in normal form.                                                                                |
| $y(t_0) = y_0$                                                                                                     | Initial condition for a scalar differential equation.                                                                                      |
| $y' + p(t)y = g(t)$                                                                                                | Standard form of a first-order linear differential equation.                                                                               |
| $\mu(t)$                                                                                                           | Integrating factor; for $y' + p(t)y = g(t)$ , one may take<br>$\mu(t) = \exp\left(\int p(t) \, dt\right)$ up to a nonzero constant factor. |
| $M(x, y) \, dx + N(x, y) \, dy = 0$                                                                                | Differential form of a first-order differential equation.                                                                                  |
| $M_x, M_y, N_x, N_y$                                                                                               | Subscript notation for the corresponding partial derivatives.                                                                              |
| $\mathcal{L}[y]$                                                                                                   | Linear differential operator applied to $y$ ; $\mathcal{L}$ is reserved here for the Laplace transform.                                    |
| $L[y] = \sum_{k=0}^n a_k(t)y^{(k)}$                                                                                | General $n$ th-order linear differential operator, with $a_n \neq 0$ .                                                                     |
| $L[y] = 0, \mathcal{L}[y] = g$                                                                                     | Homogeneous and nonhomogeneous linear differential equations, respectively.                                                                |
| $y_h, y_c$                                                                                                         | Homogeneous or complementary part of a solution; $y_h$ is preferred.                                                                       |
| $y_p$                                                                                                              | Particular solution.                                                                                                                       |
| $c_1, c_2, \dots$                                                                                                  | Arbitrary constants in a general solution.                                                                                                 |
| $\sum_{k=0}^n a_k r^k = 0$                                                                                         | Characteristic equation for a constant-coefficient homogeneous linear differential equation.                                               |
| $r_1, \dots, r_n$                                                                                                  | Characteristic roots; multiplicities are counted when relevant.                                                                            |
| $W(y_1, \dots, y_n)(t),$<br>$W[y_1, \dots, y_n](t)$                                                                | Wronskian of $y_1, \dots, y_n$ ; parentheses are preferred.                                                                                |
| $y = \sum_{n=0}^{\infty} a_n(x - x_0)^n$                                                                           | Power-series form about the ordinary point $x_0$ .                                                                                         |
| $y = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+r}$                                                                       | Frobenius-series form about a regular singular point $x_0$ ; $r$ denotes an indicial exponent.                                             |
| $\mathbf{x}' = \mathbf{f}(t, \mathbf{x})$                                                                          | General first-order system.                                                                                                                |
| $\mathbf{x}(t_0) = \mathbf{x}_0$                                                                                   | Initial condition for a first-order system.                                                                                                |
| $\mathbf{x}' = \mathbf{A}\mathbf{x} + \mathbf{g}(t),$<br>$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{g}(t)$ | Linear first-order system in state-vector form.                                                                                            |
| $\mathbf{x}' = \mathbf{f}(\mathbf{x})$                                                                             | Autonomous first-order system.                                                                                                             |
| $\mathbf{x}_e$                                                                                                     | Equilibrium point; $\mathbf{f}(\mathbf{x}_e) = \mathbf{0}$ .                                                                               |
| $\Phi(t)$                                                                                                          | Fundamental matrix of a linear system.                                                                                                     |
| $\mathcal{L}\{f(t)\}(s),$<br>$\mathcal{L}[f](s)$                                                                   | Laplace transform of $f$ .                                                                                                                 |
| $\mathcal{L}^{-1}\{F(s)\}(t),$<br>$\mathcal{L}^{-1}[F](t)$                                                         | Inverse Laplace transform of $F$ .                                                                                                         |
| $(f * g)(t)$                                                                                                       | One-sided convolution, $(f * g)(t) := \int_0^t f(\tau)g(t - \tau) \, d\tau$ .                                                              |

| Notation                                                        | Meaning                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $H(t-a), u(t-a)$                                                | Heaviside or unit-step function shifted to $t = a$ ; $H$ is preferred, and its value at the jump is specified when relevant.                                                                                                                                                                                                                                        |
| $\delta(t), \delta(t-a)$                                        | Dirac delta distribution, or unit impulse, centered at 0 or shifted to $t = a$ .                                                                                                                                                                                                                                                                                    |
| $t_n = t_0 + nh$                                                | Uniform numerical grid with step size $h > 0$ .                                                                                                                                                                                                                                                                                                                     |
| $y_n \approx y(t_n)$                                            | Numerical approximation to the exact solution at $t_n$ .                                                                                                                                                                                                                                                                                                            |
| $u_x, u_t, u_{xx}, u_{xy}, u_{tt}, \dots$                       | Subscript notation for partial derivatives of a scalar function $u$ .                                                                                                                                                                                                                                                                                               |
| $\partial_{\mathbf{n}}u$                                        | Normal derivative, $\partial_{\mathbf{n}}u := \nabla u \cdot \mathbf{n}$ , for a unit normal vector $\mathbf{n}$ .                                                                                                                                                                                                                                                  |
| $u _{\partial\Omega} = g$                                       | Dirichlet boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                                  |
| $\partial_{\mathbf{n}}u _{\partial\Omega} = g$                  | Neumann boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                                    |
| $\alpha u + \beta \partial_{\mathbf{n}}u = g$                   | Robin boundary condition on $\partial\Omega$ .                                                                                                                                                                                                                                                                                                                      |
| $a_n, b_n$                                                      | Fourier coefficients for the real $2L$ -periodic convention                                                                                                                                                                                                                                                                                                         |
|                                                                 | $f(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right).$                                                                                                                                                                                                                                               |
| $a_n$                                                           | $a_n = \frac{1}{L} \int_{-L}^L f(x) \cos \frac{n\pi x}{L} dx \text{ for } n \geq 0.$                                                                                                                                                                                                                                                                                |
| $b_n$                                                           | $b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx \text{ for } n \geq 1.$                                                                                                                                                                                                                                                                                |
| $\widehat{f}(\boldsymbol{\xi}), \mathcal{F}f(\boldsymbol{\xi})$ | Fourier transform under the normalization<br>$\widehat{f}(\boldsymbol{\xi}) := \int_{\mathbb{R}^n} f(\mathbf{x}) e^{-2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\mathbf{x}; \text{ inversion uses}$ $f(\mathbf{x}) = \int_{\mathbb{R}^n} \widehat{f}(\boldsymbol{\xi}) e^{2\pi i \mathbf{x} \cdot \boldsymbol{\xi}} d\boldsymbol{\xi} \text{ whenever applicable.}$ |
| $-(py')' + qy = \lambda wy$                                     | Sturm–Liouville eigenvalue equation, where $p, q$ , and $w$ are functions of $x$ .                                                                                                                                                                                                                                                                                  |
| $\lambda_n, \phi_n$                                             | Sturm–Liouville eigenvalue and corresponding eigenfunction.                                                                                                                                                                                                                                                                                                         |
| $\langle f, g \rangle_w$                                        | Weighted inner product, $\langle f, g \rangle_w := \int_a^b \overline{f(x)} g(x) w(x) dx$ .                                                                                                                                                                                                                                                                         |
| $\delta_{mn}$                                                   | Kronecker delta: 1 if $m = n$ and 0 otherwise.                                                                                                                                                                                                                                                                                                                      |
| $\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$                | Weighted orthonormality of eigenfunctions.                                                                                                                                                                                                                                                                                                                          |
| $G(x, \xi), G(\mathbf{x}, \boldsymbol{\xi})$                    | Green’s function; arguments are scalar or vector as appropriate.                                                                                                                                                                                                                                                                                                    |
| $\text{Ai}(x), \text{Bi}(x)$                                    | Airy functions.                                                                                                                                                                                                                                                                                                                                                     |
| $J_\nu(x)$                                                      | Bessel function of the first kind of order $\nu$ .                                                                                                                                                                                                                                                                                                                  |
| $Y_\nu(x), N_\nu(x)$                                            | Bessel function of the second kind of order $\nu$ ; $Y_\nu$ is preferred.                                                                                                                                                                                                                                                                                           |
| $I_\nu(x), K_\nu(x)$                                            | Modified Bessel functions of the first and second kinds.                                                                                                                                                                                                                                                                                                            |
| $H_\nu^{(1)}(x), H_\nu^{(2)}(x)$                                | Hankel functions of the first and second kinds.                                                                                                                                                                                                                                                                                                                     |
| $P_n(x)$                                                        | Legendre polynomial of degree $n$ .                                                                                                                                                                                                                                                                                                                                 |
| $P_\nu(x), Q_\nu(x)$                                            | Legendre functions of the first and second kinds.                                                                                                                                                                                                                                                                                                                   |
| $P_\ell^m(x)$                                                   | Associated Legendre function for integer degree $\ell$ and order $m$ .                                                                                                                                                                                                                                                                                              |
| $H_n(x), \text{He}_n(x)$                                        | Physicists’ and probabilists’ Hermite polynomials, respectively.                                                                                                                                                                                                                                                                                                    |
| $L_n^{(\alpha)}(x), L_n(x)$                                     | Generalized and ordinary Laguerre polynomials; $L_n(x) = L_n^{(0)}(x)$ .                                                                                                                                                                                                                                                                                            |
| $T_n(x), U_n(x)$                                                | Chebyshev polynomials of the first and second kinds.                                                                                                                                                                                                                                                                                                                |
| $P_n^{(\alpha, \beta)}(x)$                                      | Jacobi polynomial.                                                                                                                                                                                                                                                                                                                                                  |
| $C_n^{(\lambda)}(x)$                                            | Gegenbauer (ultraspherical) polynomial.                                                                                                                                                                                                                                                                                                                             |
| $\text{erf}(x), \text{erfc}(x)$                                 | Error function and complementary error function.                                                                                                                                                                                                                                                                                                                    |
| ${}_2F_1(a, b; c; z)$                                           | Gauss hypergeometric function.                                                                                                                                                                                                                                                                                                                                      |

## 7 Combinatorics and Discrete Mathematics

| Notation                             | Meaning                                                                                                                     |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $\binom{n}{r}, {}_nC_r$              | Number of $r$ -element subsets of an $n$ -element set.                                                                      |
| $\left(\binom{n}{r}\right), {}_nH_r$ | $r$ -combinations with repetition from $n$ types; equals $\binom{n+r-1}{r}$ .                                               |
| $n!$                                 | Factorial.                                                                                                                  |
| $n^{\underline{k}}, (n)_k$           | Falling factorial.                                                                                                          |
| $n^{\overline{k}}, n^{(k)}$          | Rising factorial.                                                                                                           |
| $O(g(n))$                            | Big-O upper asymptotic bound.                                                                                               |
| $o(g(n))$                            | Little-o strict asymptotic bound.                                                                                           |
| $\Theta(g(n))$                       | Two-sided asymptotic order.                                                                                                 |
| $\Omega(g(n))$                       | Asymptotic lower bound.                                                                                                     |
| $(X, \preceq)$                       | Partially ordered set (poset).                                                                                              |
| $\prec, \succeq, \succ$              | Strict and reverse order relations associated with $\preceq$ .                                                              |
| $\leq_{\text{lex}}$                  | Lexicographic order.                                                                                                        |
| $x \prec y$                          | Cover relation: $y$ covers $x$ .                                                                                            |
| $B_n$                                | Boolean lattice (Boolean poset) on an $n$ -element ground set.                                                              |
| $G = (V, E)$                         | Graph with vertex set $V$ and edge set $E$ .                                                                                |
| $V(G)$                               | Vertex set of $G$ .                                                                                                         |
| $E(G)$                               | Edge set of $G$ .                                                                                                           |
| $N(v), \Gamma(v)$                    | Neighborhood of vertex $v$ .                                                                                                |
| $\deg_G(v), \deg(v)$                 | Degree of vertex $v$ ; the graph subscript may be omitted when clear.                                                       |
| $\deg^+(v), \deg^-(v)$               | Out-degree and in-degree of $v$ in a directed graph.                                                                        |
| $d_G(u, v), d(u, v)$                 | Graph distance between vertices $u$ and $v$ .                                                                               |
| $K_n$                                | Complete graph on $n$ vertices.                                                                                             |
| $K_{m,n}$                            | Complete bipartite graph with parts of sizes $m$ and $n$ .                                                                  |
| $P_n$                                | Path graph on $n$ vertices.                                                                                                 |
| $C_n$                                | Cycle graph on $n$ vertices.                                                                                                |
| $\mathbf{A}(G), \mathbf{A}_G$        | Adjacency matrix of $G$ .                                                                                                   |
| $\mathbf{L}(G), \mathbf{L}_G$        | Graph Laplacian, usually $\mathbf{D}(G) - \mathbf{A}(G)$ .                                                                  |
| $G - e$                              | Graph obtained by deleting the edge $e$ .                                                                                   |
| $G/e, G : e$                         | Graph obtained by contracting the edge $e$ ; $G/e$ is the preferred notation.                                               |
| $T(G), \tau(G)$                      | Number of spanning trees of $G$ ; $T(G)$ is preferred here to avoid conflict with the standard number-theoretic $\tau(n)$ . |
| $r(k, \ell), r(k)$                   | Ramsey numbers; $r(k) = r(k, k)$ .                                                                                          |
| $A(x) = \sum_{n=0}^{\infty} a_n x^n$ | Ordinary generating function of the sequence $(a_n)_{n \geq 0}$ .                                                           |

## 8 Probability and Statistics

Probability and expectation are written in blackboard bold by default. For gamma distributions,  $\beta$  denotes a rate parameter and  $\theta$  a scale parameter, so  $\theta = 1/\beta$ . The geometric distribution counts the number of trials up to and including the first success.

| Notation              | Meaning                    |
|-----------------------|----------------------------|
| $\mathbb{P}(A), P(A)$ | Probability of event $A$ . |

| Notation                                                | Meaning                                                                                                                           |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| $\mathbb{P}(A \mid B), P(A \mid B)$                     | Conditional probability of $A$ given $B$ .                                                                                        |
| $X \perp\!\!\!\perp Y$                                  | $X$ and $Y$ are independent.                                                                                                      |
| $X \perp\!\!\!\perp Y \mid Z$                           | $X$ and $Y$ are conditionally independent given $Z$ .                                                                             |
| $p_X(x) = \mathbb{P}(X = x)$                            | Probability mass function (pmf) of a discrete random variable $X$ .                                                               |
| $f_X(x)$                                                | Probability density function (pdf) of a continuous random variable $X$ .                                                          |
| $F_X(x) = \mathbb{P}(X \leq x)$                         | Cumulative distribution function (cdf) of $X$ .                                                                                   |
| $p_{X,Y}(x, y), f_{X,Y}(x, y)$                          | Joint pmf or joint pdf of $(X, Y)$ , according to context.                                                                        |
| $p_{X Y}(x \mid y), f_{X Y}(x \mid y)$                  | Conditional pmf or conditional pdf of $X$ given $Y = y$ , according to context.                                                   |
| $F^{-1}(p)$                                             | Generalized inverse of a cdf $F$ , $F^{-1}(p) = \inf\{x \in \mathbb{R} \mid F(x) \geq p\}$ ; the $p$ -quantile, for $0 < p < 1$ . |
| $\mathbb{E}[X], E[X]$                                   | Expectation of random variable $X$ .                                                                                              |
| $\mathbb{E}[X \mid Y], E[X \mid Y]$                     | Conditional expectation of $X$ given $Y$ .                                                                                        |
| $\text{Var}(X)$                                         | Variance of $X$ .                                                                                                                 |
| $\text{Cov}(X, Y)$                                      | Covariance of $X$ and $Y$ .                                                                                                       |
| $\rho_{X,Y}$                                            | Population correlation coefficient of $X$ and $Y$ .                                                                               |
| $r$                                                     | Sample correlation coefficient, when the sample is understood.                                                                    |
| $\varphi_X(t) = \mathbb{E}[e^{itX}]$                    | Characteristic function of $X$ .                                                                                                  |
| $M_X(t) = \mathbb{E}[e^{tX}]$                           | Moment-generating function (mgf) of $X$ , when finite in a neighborhood of 0.                                                     |
| $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$                | Sample mean as a statistic of a random sample $X_1, \dots, X_n$ .                                                                 |
| $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$                | Observed sample mean for data $x_1, \dots, x_n$ .                                                                                 |
| $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$    | Sample variance as a statistic.                                                                                                   |
| $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$    | Observed sample variance.                                                                                                         |
| $S = \sqrt{S^2}, s = \sqrt{s^2}$                        | Sample standard deviation as a statistic and its observed value.                                                                  |
| $\hat{\theta}$                                          | Estimator or estimate of a parameter $\theta$ , according to context.                                                             |
| $\hat{p}$                                               | Estimator or estimate of a probability parameter $p$ ; commonly the sample proportion.                                            |
| $\text{SE}(\hat{\theta})$                               | Standard error of an estimator $\hat{\theta}$ .                                                                                   |
| $L(\theta; \mathbf{x})$                                 | Likelihood function for parameter $\theta$ given observed data $\mathbf{x}$ .                                                     |
| $\ell(\theta; \mathbf{x}) = \log L(\theta; \mathbf{x})$ | Log-likelihood function.                                                                                                          |
| $H_0$                                                   | Null hypothesis.                                                                                                                  |
| $H_1, H_a$                                              | Alternative hypothesis; $H_1$ is preferred.                                                                                       |
| $\alpha$                                                | Significance level; an upper bound on the Type I error probability under the specified test.                                      |
| $\beta$                                                 | Type II error probability under a specified alternative.                                                                          |
| $\nu$                                                   | Degrees of freedom.                                                                                                               |
| $X \sim \text{Bern}(p)$                                 | Bernoulli distribution with success probability $p$ .                                                                             |
| $X \sim \text{Bin}(n, p)$                               | Binomial distribution.                                                                                                            |
| $X \sim \text{Geom}(p),$                                | Geometric distribution on $\mathbb{N}$ , counting trials up to and including the first success.                                   |
| $\mathbb{P}(X = k) = (1-p)^{k-1}p$                      |                                                                                                                                   |
| $X \sim \text{Pois}(\lambda)$                           | Poisson distribution with parameter $\lambda$ .                                                                                   |

| Notation                                   | Meaning                                                                               |
|--------------------------------------------|---------------------------------------------------------------------------------------|
| $X \sim \text{Unif}(a, b)$                 | Continuous uniform distribution on $[a, b]$ .                                         |
| $X \sim \text{Exp}(\lambda)$               | Exponential distribution with rate $\lambda$ .                                        |
| $X \sim \Gamma(\alpha, \beta)$             | Gamma distribution with shape $\alpha$ and rate $\beta$ .                             |
| $X \sim \Gamma(\alpha, \theta)$            | Gamma distribution with shape $\alpha$ and scale $\theta$ ; here $\theta = 1/\beta$ . |
| $X \sim \mathcal{N}(\mu, \sigma^2),$       | Normal distribution with mean $\mu$ and variance $\sigma^2$ .                         |
| $X \sim N(\mu, \sigma^2)$                  |                                                                                       |
| $X \sim \chi_\nu^2$                        | Chi-squared distribution with $\nu$ degrees of freedom.                               |
| $X \sim t_\nu$                             | Student's $t$ distribution with $\nu$ degrees of freedom.                             |
| $X \sim F_{\nu_1, \nu_2}$                  | $F$ distribution with numerator and denominator degrees of freedom $\nu_1, \nu_2$ .   |
| $X_n \xrightarrow{\text{a.s.}} X$          | Almost-sure convergence.                                                              |
| $X_n \xrightarrow{\mathbb{P}} X$           | Convergence in probability.                                                           |
| $X_n \xrightarrow{L^p} X$                  | Convergence in $L^p$ .                                                                |
| $X_n \xrightarrow{d} X, X_n \Rightarrow X$ | Convergence in distribution.                                                          |

## 9 Geometry

In triangle notation,  $A, B, C$  denote vertices and  $a, b, c$  the opposite side lengths.

| Notation                    | Meaning                                |
|-----------------------------|----------------------------------------|
| $\triangle ABC$             | Triangle with vertices $A, B, C$ .     |
| $\angle ABC$                | Angle with vertex $B$ .                |
| $\overline{AB}$             | Segment joining $A$ and $B$ .          |
| $AB$                        | Length of $\overline{AB}$ .            |
| $[ABC]$                     | Area of $\triangle ABC$ .              |
| $\text{length}(C), \ell(C)$ | Length of $C$ .                        |
| $\text{area}(S)$            | Area of $S$ .                          |
| $\text{vol}(S)$             | Volume of $S$ .                        |
| $\text{dist}(P, S)$         | Distance from $P$ to $S$ .             |
| $\text{conv}(S)$            | Convex hull of $S$ .                   |
| $\mathbb{R}^n$              | $n$ -dimensional real Euclidean space. |
| $\mathbb{H}^n$              | $n$ -dimensional hyperbolic space.     |
| $\parallel$                 | Parallel.                              |
| $\perp$                     | Perpendicular or orthogonal.           |

## 10 Number Theory

| Notation                          | Meaning                                     |
|-----------------------------------|---------------------------------------------|
| $a \mid b$                        | $a$ divides $b$ .                           |
| $a \nmid b$                       | $a$ does not divide $b$ .                   |
| $\gcd(a, b)$                      | Greatest common divisor, taken nonnegative. |
| $\text{lcm}(a, b)$                | Least common multiple.                      |
| $a \equiv b \pmod{n}$             | $a$ is congruent to $b$ modulo $n$ .        |
| $[a]_n$                           | Residue class of $a$ modulo $n$ .           |
| $\mathbb{Z}/n\mathbb{Z}$          | Ring of residue classes modulo $n$ .        |
| $(\mathbb{Z}/n\mathbb{Z})^\times$ | Multiplicative group of units modulo $n$ .  |

| Notation                     | Meaning                                         |
|------------------------------|-------------------------------------------------|
| $a^{-1} \pmod n$             | Multiplicative inverse of $a$ modulo $n$ .      |
| $\left(\frac{a}{p}\right)$   | Legendre symbol.                                |
| $\varphi(n)$                 | Euler totient function.                         |
| $\text{ord}_n(a)$            | Multiplicative order of $a$ modulo $n$ .        |
| $v_p(n), \nu_p(n)$           | $p$ -adic valuation of $n$ .                    |
| $\tau(n)$                    | Number of positive divisors of $n$ .            |
| $\sigma(n)$                  | Sum of positive divisors of $n$ .               |
| $\mu(n)$                     | Möbius function.                                |
| $\text{rad}(n)$              | Product of the distinct prime divisors of $n$ . |
| $\pi(x)$                     | Number of primes not exceeding $x$ .            |
| $\mathbb{F}_q, \text{GF}(q)$ | Finite field with $q$ elements.                 |

## 11 Algebra

$G, H$  usually denote groups,  $R, S$  rings,  $I, J$  ideals,  $M, N$  modules, and  $F, K, L$  fields. Modules are left modules unless otherwise stated.

| Notation                   | Meaning                                              |
|----------------------------|------------------------------------------------------|
| $G$                        | Group.                                               |
| $e, 1_G$                   | Identity element of a group.                         |
| $g^{-1}$                   | Inverse of $g \in G$ .                               |
| $ G $                      | Order (cardinality) of $G$ .                         |
| $\text{ord}(g),  g $       | Order of the element $g$ .                           |
| $\langle S \rangle$        | Subgroup generated by $S \subseteq G$ .              |
| $\langle g \rangle$        | Cyclic subgroup generated by $g$ .                   |
| $H \leq G$                 | $H$ is a subgroup of $G$ .                           |
| $H \trianglelefteq G$      | $H$ is a normal subgroup of $G$ .                    |
| $gH, Hg$                   | Left and right cosets of $H$ in $G$ .                |
| $[G : H]$                  | Index of $H$ in $G$ .                                |
| $G/H$                      | Quotient group, for $H \trianglelefteq G$ .          |
| $\ker \varphi$             | Kernel of the homomorphism $\varphi$ .               |
| $\text{im } \varphi$       | Image of the homomorphism $\varphi$ .                |
| $\text{Aut}(G)$            | Automorphism group of $G$ .                          |
| $\text{Inn}(G)$            | Inner automorphism group of $G$ .                    |
| $Z(G)$                     | Center of $G$ .                                      |
| $C_G(A), C_G(x)$           | Centralizer in $G$ of $A \subseteq G$ or $x \in G$ . |
| $N_G(A), N_G(H)$           | Normalizer in $G$ of $A \subseteq G$ or $H \leq G$ . |
| $[x, y] := x^{-1}y^{-1}xy$ | Commutator of $x$ and $y$ .                          |
| $[G, G], G'$               | Commutator (derived) subgroup of $G$ .               |
| $\text{Syl}_p(G)$          | Set of Sylow $p$ -subgroups of $G$ .                 |
| $n_p$                      | Number of Sylow $p$ -subgroups of $G$ .              |
| $G \curvearrowright X$     | Action of $G$ on $X$ .                               |
| $g \cdot x$                | Action of $g \in G$ on $x \in X$ .                   |
| $G \cdot x$                | Orbit of $x$ under the action of $G$ .               |
| $\text{Stab}_G(x), G_x$    | Stabilizer of $x$ in $G$ .                           |
| $G \times H$               | Direct product of groups.                            |

| Notation                                  | Meaning                                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------|
| $N \rtimes_{\varphi} H$                   | Semidirect product determined by $\varphi: H \rightarrow \text{Aut}(N)$ , with $N$ the normal factor. |
| $\langle S \mid R \rangle$                | Group presentation with generators $S$ and relations $R$ .                                            |
| $C_n$                                     | Cyclic group of order $n$ .                                                                           |
| $D_n$                                     | Dihedral group of symmetries of a regular $n$ -gon; $ D_n  = 2n$ .                                    |
| $S_n, \mathfrak{S}_n$                     | Symmetric group on $n$ symbols.                                                                       |
| $A_n, \mathfrak{A}_n$                     | Alternating group on $n$ symbols.                                                                     |
| $V_4$                                     | Klein four-group.                                                                                     |
| $Q_8$                                     | Quaternion group of order 8.                                                                          |
| $\text{GL}_n(F)$                          | General linear group over $F$ .                                                                       |
| $\text{SL}_n(F)$                          | Special linear group over $F$ .                                                                       |
| $R$                                       | Ring.                                                                                                 |
| $0_R, 1_R$                                | Additive and multiplicative identities of $R$ , when disambiguation is useful.                        |
| $R^{\times}$                              | Group of units of $R$ .                                                                               |
| $I, J$                                    | Ideals of $R$ , when the ambient ring is clear.                                                       |
| $\mathfrak{p}, \mathfrak{q}$              | Prime ideals.                                                                                         |
| $\mathfrak{m}, \mathfrak{n}$              | Maximal ideals.                                                                                       |
| $(a)$                                     | Principal ideal generated by $a$ .                                                                    |
| $(a_1, \dots, a_n)$                       | Ideal generated by $a_1, \dots, a_n$ .                                                                |
| $I + J$                                   | Sum of the ideals $I$ and $J$ .                                                                       |
| $IJ$                                      | Product of the ideals $I$ and $J$ .                                                                   |
| $R/I$                                     | Quotient ring.                                                                                        |
| $R[x], R[x_1, \dots, x_n]$                | Polynomial rings over $R$ in one or several variables.                                                |
| $\deg f$                                  | Degree of the polynomial $f$ .                                                                        |
| $M_n(R)$                                  | Ring of $n \times n$ matrices over $R$ .                                                              |
| $R[G]$                                    | Group ring of $G$ over $R$ .                                                                          |
| $\text{Frac}(R)$                          | Fraction field of the integral domain $R$ .                                                           |
| $\text{char } R$                          | Characteristic of $R$ .                                                                               |
| $\mathbb{H}$                              | Quaternion division algebra over $\mathbb{R}$ .                                                       |
| $M, N$                                    | Left $R$ -modules, unless otherwise stated.                                                           |
| ${}_R M, M_R$                             | Left and right $R$ -modules, when the side is emphasized.                                             |
| $M/N$                                     | Quotient module, for a submodule $N$ of $M$ .                                                         |
| $M \oplus N$                              | Direct sum of modules.                                                                                |
| $\bigoplus_{i \in I} M_i$                 | Direct sum of an indexed family of modules.                                                           |
| $R^n$                                     | Free $R$ -module of rank $n$ .                                                                        |
| $\text{Hom}_R(M, N)$                      | Set of $R$ -module homomorphisms from $M$ to $N$ .                                                    |
| $\text{End}_R(M)$                         | Endomorphism ring of the $R$ -module $M$ .                                                            |
| $\text{Aut}_R(M)$                         | Automorphism group of the $R$ -module $M$ .                                                           |
| $\text{Ann}_R(M)$                         | Annihilator of the $R$ -module $M$ .                                                                  |
| $M \otimes_R N$                           | Tensor product over $R$ , with the required right and left module structures understood.              |
| $K/F$                                     | Field extension of $F$ by $K$ .                                                                       |
| $[K : F]$                                 | Degree of the field extension $K/F$ .                                                                 |
| $F(\alpha), F(\alpha_1, \dots, \alpha_n)$ | Fields generated over $F$ by the indicated elements.                                                  |
| $F[\alpha]$                               | Subring (equivalently, $F$ -subalgebra) generated by $F$ and $\alpha$ .                               |
| $m_{\alpha, F}(x)$                        | Minimal polynomial of $\alpha$ over $F$ .                                                             |

| Notation                  | Meaning                                         |
|---------------------------|-------------------------------------------------|
| $\overline{F}$            | Algebraic closure of $F$ .                      |
| $\text{Gal}(L/K)$         | Galois group of $L/K$ .                         |
| $L^H$                     | Fixed field of $H \leq \text{Gal}(L/K)$ .       |
| $\text{Tr}_{L/K}(\alpha)$ | Field trace of $\alpha \in L$ from $L$ to $K$ . |
| $N_{L/K}(\alpha)$         | Field norm of $\alpha \in L$ from $L$ to $K$ .  |
| $\text{trdeg}_K L$        | Transcendence degree of $L$ over $K$ .          |

## 12 Topology

| Notation                        | Meaning                                             |
|---------------------------------|-----------------------------------------------------|
| $(X, \mathcal{T})$              | Topological space $X$ with topology $\mathcal{T}$ . |
| $A^\circ$ , $\text{int } A$     | Interior of $A$ .                                   |
| $\overline{A}$ , $\text{cl } A$ | Closure of $A$ .                                    |
| $\partial A$                    | Boundary of $A$ .                                   |
| $\pi_1(X, x_0)$                 | Fundamental group of $X$ based at $x_0$ .           |
| $\chi(X)$                       | Euler characteristic of $X$ .                       |
| $X \cong Y$                     | Homeomorphic spaces.                                |
| $X \simeq Y$                    | Homotopy equivalent spaces.                         |

## 13 Miscellaneous Notation

| Notation            | Meaning                                                     |
|---------------------|-------------------------------------------------------------|
| $\lfloor x \rfloor$ | Floor of $x$ .                                              |
| $\lceil x \rceil$   | Ceiling of $x$ .                                            |
| $\{x\}$             | Fractional part $x - \lfloor x \rfloor$ , when unambiguous. |
| $\text{sgn}(x)$     | Sign of a real number or permutation, according to context. |
| $\Delta q$          | Finite change in a quantity $q$ .                           |
| $\epsilon_{ijk}$    | Levi-Civita symbol.                                         |